# Tri-Level Calibration for Training-Free Few-Shot Class-Incremental Learning with Label-Free Query Prototype Rectification

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**Abstract**

Few-shot class-incremental learning (FSCIL) must learn new classes from a handful of examples per class without forgetting the abundant base classes seen during pre-training. To sidestep catastrophic forgetting, recent training-free methods build a bi-level mixture of a CLIP visual prototype and a CLIP text prototype per class. This strategy hits two limits: CLIP's image embeddings pack visually similar classes too tightly to separate fine-grained novel ones, and the noisy five-shot novel prototypes have no mechanism for refinement after deployment. We propose Tri-Level Calibration (TRIC), a training-free framework that separates the combination into three calibration steps: a semantic calibration inside the language branch, an inter-encoder calibration that fuses CLIP and DINOv2 visual evidence on a common score axis, and an inter-modal calibration between the resulting visual and semantic evidence. On top of this calibrated backbone we introduce Query Prototype Rectification (QPR), a label-free module that uses the unlabeled query batch to refine only the noisy novel visual prototypes. Across three FSCIL benchmarks, TRIC improves all-session accuracy over strong frozen-encoder baselines with positive gains in all 20 seeds, and QPR adds further gains without any additional annotation.

## 1 Introduction

Few-shot class-incremental learning (FSCIL) evaluates recognition under a challenging protocol: a strong base classifier is learned first, and each later session introduces new classes with only a handful of labeled support images [24, 30, 32, 33]. Frozen FSCIL methods avoid incremental gradient updates altogether and therefore sidestep catastrophic forgetting [8, 15, 19, 20]. The cost is that the final classifier must be assembled at inference time from imperfect fixed evidence: language priors, five-shot visual prototypes, and the unlabeled query batch. The recent BiMC framework [8] formalizes this assembly as *bi-level modality calibration*: CLIP semantic prototypes are first enriched with class descriptions, then combined with CLIP visual prototypes through a soft base-prototype shift, and finally fused with auxiliary base covariance and novel description-retrieval evidence.

This view is internally consistent when the visual signal comes entirely from CLIP, but two structural issues surface as soon as one looks beyond a single visual encoder. The first is geometric. Adding a self-supervised visual encoder such as DINOv2 [18] introduces a second visual space that is qualitatively different from CLIP's: CLIP's image branch is

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trained to align with text and produces visually dense class prototypes, while DINOv2 is trained for visual self-similarity and produces much sparser ones. The two embedding spaces are not commensurable, so averaging their prototypes is geometrically meaningless, and treating them with a single mixing coefficient discards the very structural difference that makes DINOv2 useful for separating fine-grained novel classes. The second is statistical. Every novel prototype is still estimated from only five labeled images, so a non-trivial fraction of novel centers remain noisy even when the calibrated posterior is otherwise well-shaped. This noise is not purely a function of session age; on fine-grained benchmarks it tracks local cluster density instead, so any refinement that targets it must be precision-favoring rather than late-session-favoring. Its only available supervision source beyond the labeled support is the *unlabeled* query batch, used through a conservative gate that keeps base-class queries out of novel prototypes.

The right generalization is therefore not a different mixture coefficient on top of BiMC's bi-level form, but a recognition that frozen FSCIL is a *three*-level calibration problem: (i) calibrate semantic evidence within the language branch, (ii) calibrate the two visual encoders against each other on a common class-score axis (a per-query vector of cosine similarities indexed by the observed class set), and only then (iii) calibrate the resulting visual evidence against the semantic evidence. The first two levels reshape the calibrated backbone itself; the residual five-shot prototype noise is a failure mode that no amount of re-mixing can repair, and is addressed by a separate label-free rectification stage that operates on top of the calibrated logits.

**Contributions.**

- We reframe frozen FSCIL as a *tri-level* calibration problem, separating semantic, inter-model visual, and inter-modal calibration, and we show empirically that this separation is required once a second visual encoder is introduced.
- We propose QPR as a downstream label-free rectification module that updates only DINOv2 novel prototypes through a precision-favoring gate driven by the calibrated posterior.
- Over a 20-seed evaluation on three FSCIL benchmarks, TRIC alone exceeds the strongest training-free baseline by $+9.83$, $+1.32$, and $+12.53$ points, and TRIC-QPR adds further label-free gains with positive improvement in every seed.

## 2 Related Work

**Few-shot class-incremental learning.** FSCIL extends class-incremental learning to a low-shot regime [24]. It inherits the forgetting problem [8, 15, 19, 21] and depends on few-shot generalization [9, 10, 23, 25]. Trainable approaches evolve classifiers incrementally [30], distill semantic information [6], learn forward-compatible representations [33], meta-learn adaptation paths [7], or impose prototype-geometric regularizers [32]. Our method introduces no trainable incremental module; it operates entirely at inference time on frozen encoders.

**Frozen FSCIL and vision-language adaptation.** Training-free FSCIL methods avoid incremental retraining altogether. TEEN calibrates novel prototypes via base statistics [28], and BiMC formulates frozen FSCIL as bi-level modality calibration on top of CLIP semantic

and CLIP visual prototypes [5]. Prompt and adapter methods provide complementary evidence that frozen vision-language backbones can be improved without retraining [12, 22, 31, 34]. Our work changes the *structural form* of the calibration. Concretely, where BiMC mixes CLIP semantic and CLIP visual prototypes with a soft base-prototype shift and a class-type-specific auxiliary ensemble (a base covariance branch and a novel description-nearest-neighbor branch), TRIC introduces an inter-model visual calibration that operates on class-score vectors rather than on embeddings, drops the soft shift because it injects base bias into the DINOv2 visual carrier, and removes the auxiliary ensemble entirely so that a single calibrated posterior remains. This single posterior is what QPR consumes as a clean gate signal.

**Self-supervised features and transduction.** DINOv2 sits in a broader self-supervised visual-representation literature [2, 3, 11, 18], complementary to supervised vision transformers [9] and vision-language pretraining [14, 20]. Transductive few-shot learning and test-time adaptation use the unlabeled query batch as additional signal [1, 16, 17, 27, 29]. In frozen FSCIL the query batch is contaminated by strong base classes, so the update rule must prefer precision over coverage; this motivates the conservative gating of our QPR stage.

# 3 Method

## 3.1 Problem Setting

We follow standard FSCIL [24]. A stream of sessions $\{\mathcal{D}^{(0)}, \mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(T)}\}$, where $T$ is the number of incremental sessions, has mutually disjoint class sets $\{\mathcal{C}^{(0)}, \mathcal{C}^{(1)}, \ldots, \mathcal{C}^{(T)}\}$. The base session $\mathcal{D}^{(0)}$ contains many labeled examples, while each incremental session $\mathcal{D}^{(t)}$, $t \geq 1$, provides $N$ new classes with $K$ labeled support images per class. At session $t$ the observed label space, base-class set, and accumulated novel-class set are

$$\mathcal{C}_{\leq t} = \bigcup_{s=0}^{t} \mathcal{C}^{(s)}, \qquad \mathcal{C}_b = \mathcal{C}^{(0)}, \qquad \mathcal{C}_n = \mathcal{C}_{\leq t} \setminus \mathcal{C}_b. \tag{1}$$

Thus $\mathcal{C}_n$ denotes all incremental classes observed up to session $t$, not only the classes introduced in the last session. The support set of class $c$ is $\mathcal{X}_c = \{\boldsymbol{x}_{c,j}\}_{j=1}^{m_c}$, where $m_c$ is large for $c \in \mathcal{C}_b$ and $m_c = K$ for $c \in \mathcal{C}_n$. Previous sessions' training images are not revisited; after session $t$ the model classifies an unlabeled query batch $\mathcal{U}_t$ whose labels belong to $\mathcal{C}_{\leq t}$.

## 3.2 Overview

BiMC [5] formalizes frozen FSCIL as bi-level modality calibration on a single CLIP backbone: it builds a CLIP semantic prototype and a CLIP visual prototype per class, combines them through a soft base-prototype shift followed by a semantic–visual mixture, and adds a masked auxiliary ensemble (a base-covariance branch for base queries and a novel description-nearest-neighbour branch for novel queries). This bi-level form leaves no room for a second, geometrically distinct visual encoder and no mechanism to refine the noisy five-shot novel prototypes after deployment. We address both. Tri-Level Calibration (TRIC) replaces the bi-level mixture with three explicit calibration steps: semantic, inter-model visual, and inter-modal. So, a self-supervised encoder enters without averaging incompatible embedding spaces; Query Prototype Rectification (QPR) is a label-free refinement of the novel DINOv2

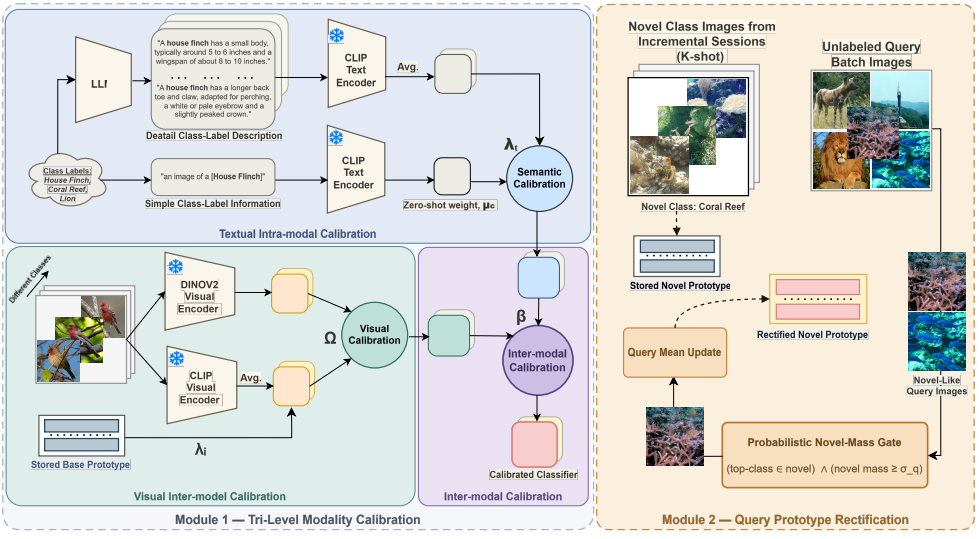



Figure 1: **Overview of TRIC-QPR.** TRIC breaks down into semantic, inter-model visual, and inter-modal calibration. QPR refines only novel visual prototypes through a precision-favoring gate.

prototypes that consumes TRIC's calibrated posterior as a gate signal. The two stages run in supervision-strength order: the labeled-support-only operations (TRIC) come before the unlabeled-query operation (QPR), so the gate QPR consumes is shaped by a posterior the gate did not produce.

## 3.3 Multi–Modal Prototypes

**Encoder notation.** Let $f_I$ and $f_T$ be the frozen CLIP image and text encoders, and let $f_D$ be the frozen DINOv2 image encoder. For any nonzero vector $\boldsymbol{u}$, $\mathrm{norm}(\boldsymbol{u})$ denotes $\ell_2$ normalization. We use unit-normalized query and support features throughout; for any image $\boldsymbol{x}$, $q_C(\boldsymbol{x})$ and $q_D(\boldsymbol{x})$ denote its CLIP-image and DINOv2 features:

$$\mathrm{norm}(\boldsymbol{u}) = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|_2}, \qquad q_C(\boldsymbol{x}) = \mathrm{norm}(f_I(\boldsymbol{x})), \qquad q_D(\boldsymbol{x}) = \mathrm{norm}(f_D(\boldsymbol{x})). \tag{2}$$

**Semantic inputs.** For each class $c \in \mathcal{C}_{\le t}$, we use a class prompt $\boldsymbol{\pi}_c$ and $n_c$ description sentences $\{d_{c,r}\}_{r=1}^{n_c}$ drawn from BiMC's prompt template and description pool (Sec. 4.1). The prompt embedding $w_c$, each description embedding $z_{c,r}$, and the normalized description mean $\bar{z}_c$ are

$$w_c = \mathrm{norm}(f_T(\boldsymbol{\pi}_c)), \qquad z_{c,r} = \mathrm{norm}(f_T(d_{c,r})), \qquad \bar{z}_c = \mathrm{norm}\left(\frac{1}{n_c}\sum_{r=1}^{n_c} z_{c,r}\right). \tag{3}$$

**Visual prototypes.** For each observed class we build one CLIP visual prototype and one DINOv2 visual prototype from the same support set $\mathcal{X}_c$:

$$v_{C,c} = \mathrm{norm}\left(\frac{1}{m_c}\sum_{j=1}^{m_c} q_C(\boldsymbol{x}_{c,j})\right), \qquad v_{D,c} = \mathrm{norm}\left(\frac{1}{m_c}\sum_{j=1}^{m_c} q_D(\boldsymbol{x}_{c,j})\right), \qquad c \in \mathcal{C}_{\le t}. \tag{4}$$

## 3.4 Tri-Level Calibration (TRIC)

**Level 1: Semantic calibration.** The prompt prototype $w_c$ and description mean $\bar{z}_c$ are combined before any visual evidence is used. The coefficient $\lambda_T$ controls how much description information enters the resulting semantic prototype $s_c$:

$$s_c = \mathrm{norm}((1-\lambda_T)w_c + \lambda_T \bar{z}_c), \qquad 0 \le \lambda_T \le 1. \tag{5}$$

For a query image $\boldsymbol{x}$, the semantic score of class $c$ is the cosine similarity between the CLIP image embedding and the semantic prototype which is well defined because CLIP's joint image–text training aligns both into one space:

$$\ell_c^S(\boldsymbol{x}) = q_C(\boldsymbol{x})^\top s_c, \qquad c \in \mathcal{C}_{\le t}. \tag{6}$$

**Level 2: Inter-model visual calibration.** The CLIP-image and DINOv2 branches first produce separate *class-score vectors* which are vectors of cosine similarities indexed by $\mathcal{C}_{\le t}$ and are then fused along this common axis rather than averaged in either encoder's embedding space. For query image $\boldsymbol{x}$ and class $c$, $\ell_c^C(\boldsymbol{x})$ and $\ell_c^D(\boldsymbol{x})$ denote the CLIP-image and DINOv2 visual logits, and $\ell_c^V(\boldsymbol{x})$ denotes the fused visual logit. Both branch logits are bounded cosine similarities in $[-1,1]$, so no per-branch rescaling is required before fusion. The coefficient $\omega$ weights DINOv2 visual evidence against CLIP visual evidence:

$$\begin{aligned} \ell_c^C(\boldsymbol{x}) &= q_C(\boldsymbol{x})^\top v_{C,c}, & \ell_c^D(\boldsymbol{x}) &= q_D(\boldsymbol{x})^\top v_{D,c}, \\ \ell_c^V(\boldsymbol{x}) &= \omega \ell_c^D(\boldsymbol{x}) + (1-\omega)\ell_c^C(\boldsymbol{x}), & 0 \le \omega \le 1. \end{aligned} \tag{7}$$

**Level 3: Inter-modal calibration.** Semantic and visual evidence are then fused at the logit level. For class $c$, $\ell_c^F(\boldsymbol{x})$ denotes the fused semantic-visual logit. Here $\beta$ weights semantic evidence against visual evidence, and $\rho$ is the positive softmax scale used to convert fused logits into the TRIC posterior $p_{\mathrm{TRIC}}(c \mid \boldsymbol{x})$:

$$\begin{aligned} \ell_c^F(\boldsymbol{x}) &= \beta \ell_c^S(\boldsymbol{x}) + (1-\beta)\ell_c^V(\boldsymbol{x}), \\ p_{\mathrm{TRIC}}(c \mid \boldsymbol{x}) &= \frac{\exp(\rho\, \ell_c^F(\boldsymbol{x}))}{\sum_{k \in \mathcal{C}_{\le t}} \exp(\rho\, \ell_k^F(\boldsymbol{x}))}, \qquad 0 \le \beta \le 1,\ \rho > 0. \end{aligned} \tag{8}$$

## 3.5 Query Prototype Rectification (QPR)

After TRIC produces the calibrated posterior $p^\star(c \mid \boldsymbol{x}) = p_{\mathrm{TRIC}}(c \mid \boldsymbol{x})$, each novel DINOv2 prototype is still estimated from $K$ images. The unlabeled query batch $\mathcal{U}_t$ contains additional structure, but it also contains base-class queries. QPR therefore uses $p^\star$ only to select high-precision novel-like queries and updates only novel DINOv2 prototypes. We use $\tau_q \in [0,1]$ as the minimum total novel posterior mass required to admit a query, and $m_{\min} \in \mathbb{N}$ as the minimum number of admitted queries required to update a class.

**Gate.** For each query $\boldsymbol{x} \in \mathcal{U}_t$, define its full prediction $\hat{y}(\boldsymbol{x})$, its best novel-class prediction $\hat{y}_n(\boldsymbol{x})$, and its total novel posterior mass $m_n(\boldsymbol{x})$:

$$\hat{y}(\boldsymbol{x}) = \arg\max_{k \in \mathcal{C}_{\le t}} p^\star(k \mid \boldsymbol{x}), \qquad \hat{y}_n(\boldsymbol{x}) = \arg\max_{k \in \mathcal{C}_n} p^\star(k \mid \boldsymbol{x}), \qquad m_n(\boldsymbol{x}) = \sum_{k \in \mathcal{C}_n} p^\star(k \mid \boldsymbol{x}). \tag{9}$$

The global gate $\mathcal{G}$ contains queries whose top class is novel and whose total novel mass exceeds $\tau_q$. The class-specific pool $\mathcal{G}_c$ then assigns each admitted query to its best novel class:

$$\mathcal{G} = \{\boldsymbol{x} \in \mathcal{U}_t : \hat{y}(\boldsymbol{x}) \in \mathcal{C}_n \text{ and } m_n(\boldsymbol{x}) \ge \tau_q\}, \qquad \mathcal{G}_c = \{\boldsymbol{x} \in \mathcal{G} : \hat{y}_n(\boldsymbol{x}) = c\}, \quad c \in \mathcal{C}_n. \tag{10}$$

If $|\mathcal{G}_c| < m_{\min}$, class $c$ is not updated and its prototype remains $v_{D,c}$.

**Update.** A probability-weighted DINOv2 query mean $\tilde{v}_{D,c}$ is computed only for novel classes with $|\mathcal{G}_c| \ge m_{\min}$. For such a class, $a_c(\boldsymbol{x})$ is a normalized weight over the gated pool, so $\sum_{\boldsymbol{x} \in \mathcal{G}_c} a_c(\boldsymbol{x}) = 1$:

$$a_c(\boldsymbol{x}) = \frac{p^\star(c \mid \boldsymbol{x})}{\sum_{\boldsymbol{x}' \in \mathcal{G}_c} p^\star(c \mid \boldsymbol{x}')}, \qquad \tilde{v}_{D,c} = \operatorname{norm}\left( \sum_{\boldsymbol{x} \in \mathcal{G}_c} a_c(\boldsymbol{x}) q_D(\boldsymbol{x}) \right). \tag{11}$$

The final DINOv2 prototype $v'_{D,c}$ is a convex interpolation between the original support prototype and the query mean for updated novel classes, where $\alpha$ controls the query-mean contribution; all other prototypes are unchanged:

$$v'_{D,c} = \begin{cases} \operatorname{norm}((1-\alpha) v_{D,c} + \alpha \tilde{v}_{D,c}), & c \in \mathcal{C}_n \text{ and } |\mathcal{G}_c| \ge m_{\min}, \\ v_{D,c}, & \text{otherwise}, \end{cases} \qquad 0 \le \alpha \le 1. \tag{12}$$

CLIP features, semantic prototypes, CLIP visual prototypes, base DINOv2 prototypes, and all encoder weights remain frozen. The rectified classifier is obtained by replacing $v_{D,c}$ with $v'_{D,c}$ in Eq. 7 and then recomputing Eq. 8.

# 4 Experiments

## 4.1 Setup

**Datasets and protocol.** We use the standard FSCIL protocols [6, 24]. CIFAR-100 [13] and *mini*ImageNet [25] use 60 base classes and eight 5-way 5-shot incremental sessions. CUB-200-2011 [26] uses 100 base classes and ten 10-way 5-shot incremental sessions. We report final-session accuracy over all observed classes, averaged over 20 seeds per configuration.

**Implementation.** All variants use CLIP ViT-B/16 [20]; TRIC additionally uses DINOv2 ViT-B/14 [18], which is comparable in FLOPs to the CLIP image encoder, so the dominant inference cost of TRIC-QPR is approximately twice that of CLIP-only frozen FSCIL methods. Class descriptions and prompt templates follow BiMC [5]. We fix $\lambda_T = 0.5$ and disable BiMC's soft visual calibration, base covariance branch, description nearest-neighbour branch, and masked auxiliary ensemble. For TRIC, $\omega = 0.7$ and $\rho = 100$. Following BiMC's convention of allowing one semantic–visual mixing coefficient per dataset, the semantic–visual weight is $\beta = 0.65$ on CIFAR-100 and *mini*ImageNet and $\beta = 0.30$ on CUB-200; all other hyperparameters are dataset-agnostic. For QPR, $\tau_q = 0.3$, $m_{\min} = 5$, and $\alpha = 0.75$, fixed across both datasets and seeds. No per-seed tuning is used.

| Method | Modality | Accuracy in each session (%) ↑ | | | | | | | | | Avg. ↑ | PD ↓ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $A_0$ | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $A_5$ | $A_6$ | $A_7$ | $A_8$ | | |
| CLIP zero-shot | L | 91.27 | 91.25 | 89.74 | 89.43 | 88.98 | 88.42 | 86.96 | 86.64 | 86.15 | 88.76 | 5.12 |
| Visual prototype (CLIP) | V | 92.58 | 91.88 | 89.37 | 88.24 | 88.42 | 87.86 | 86.33 | 86.16 | 86.09 | 88.55 | 6.49 |
| TEEN | V | 92.58 | 92.06 | 89.69 | 88.60 | 88.80 | 88.31 | 86.84 | 86.71 | 86.66 | 88.92 | 5.92 |
| FeCAM | V | 94.42 | 93.71 | 91.61 | 90.59 | 90.62 | 90.22 | 88.43 | 88.17 | 88.20 | 90.66 | 6.22 |
| BiMC | V-L | 94.90 | 94.80 | 93.21 | 92.91 | 92.88 | 92.60 | 91.78 | 91.88 | 91.81 | 92.97 | 3.09 |
| BiMC Ens. | V-L | 95.47 | 95.34 | 93.80 | 93.63 | 93.51 | 93.18 | 92.51 | 92.56 | 92.40 | 93.60 | 3.07 |
| CLIP visual only | V | 92.58 | 91.86 | 89.61 | 88.82 | 88.75 | 88.16 | 86.93 | 86.82 | 86.79 | 88.92 | 5.79 |
| DINO visual only | V | 96.42 | 96.02 | 94.47 | 94.38 | 94.43 | 93.49 | 92.30 | 92.07 | 91.90 | 93.94 | 4.52 |
| TRIC | V-L | 97.25 | 97.16 | 95.85 | 95.83 | 95.87 | 95.42 | 94.62 | 94.47 | 94.34 | 95.65 | 2.91 |
| TRIC-QPR | V-L | 97.25 | 97.17 | 95.90 | 95.91 | 95.95 | 95.59 | 94.91 | 94.79 | 94.66 | 95.79 | 2.60 |

Table 1: Session-wise comparison on *mini*ImageNet. V, L, and V-L denote visual, language, and vision-language modalities; $A_t$ is accuracy at session $t$, Avg. is mean session accuracy, PD is base-to-final degradation.

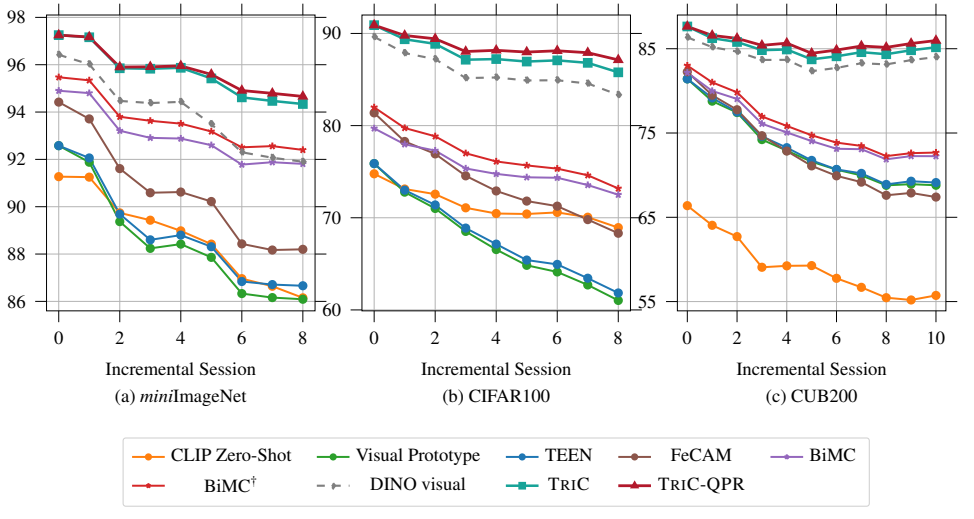



Figure 2: Performance curve of each incremental session on (a) *mini*ImageNet, (b) CIFAR100, and (c) CUB200 datasets. BiMC baselines use the reported main/supplementary values; DINOv2 visual, TRIC, and TRIC-QPR are our session-wise results.

## 4.2 Main Results

With no incremental training, the TRIC backbone improves on the BiMC ensemble baseline (BiMC with its base-covariance and description-nearest-neighbour ensemble [5]) by +9.83, +1.32, and +12.53 final-session points (73.24→83.07 on CIFAR-100, 92.41→93.73 on *mini*ImageNet, 72.76→85.29 on CUB-200), and the label-free QPR stage adds a further +2.43, +1.01, and +0.88 points on top of the same backbone, reaching 85.50, 94.74, and 86.17. TRIC-QPR beats BiMC in all 20 seeds on every benchmark, and the gains do not come at the price of base-class confusion (gate-level base leakage is reported in Table 4).

Fig. 2 shows that this is not a final-session artifact: TRIC and TRIC-QPR dominate BiMC throughout the incremental schedule on all three datasets. Table 1 gives the per-session breakdown on *mini*ImageNet; the corresponding CIFAR-100 and CUB-200 session tables are in the supplementary material.

| Components | | | *mini*ImageNet accuracy in each session (%) ↑ | | | | | | | | | Avg. ↑ | PD ↓ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sem. | C-Vis. | D-Vis. | $A_0$ | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $A_5$ | $A_6$ | $A_7$ | $A_8$ | | |
| ✓ | | | 93.22 | 93.18 | 91.79 | 91.40 | 91.18 | 90.69 | 89.32 | 89.40 | 89.26 | 91.05 | 3.96 |
| | ✓ | | 92.58 | 91.86 | 89.61 | 88.82 | 88.75 | 88.16 | 86.93 | 86.82 | 86.79 | 88.92 | 5.79 |
| | | ✓ | 96.42 | 96.02 | 94.47 | 94.38 | 94.43 | 93.49 | 92.30 | 92.07 | 91.90 | 93.94 | 4.52 |
| ✓ | ✓ | | 94.90 | 94.80 | 93.21 | 92.91 | 92.88 | 92.60 | 91.78 | 91.88 | 91.81 | 92.97 | 3.09 |
| ✓ | | ✓ | 97.06 | 96.95 | 95.67 | 95.66 | 95.70 | 95.19 | 94.37 | 94.24 | 94.09 | 95.44 | 2.97 |
| | ✓ | ✓ | 96.77 | 96.42 | 94.82 | 94.73 | 94.78 | 93.95 | 92.83 | 92.59 | 92.45 | 94.37 | 4.32 |
| ✓ | ✓ | ✓ | 97.25 | 97.16 | 95.85 | 95.83 | 95.87 | 95.42 | 94.62 | 94.47 | 94.34 | 95.65 | 2.91 |

Table 2: Component ablation of TRIC on *mini*ImageNet. Sem., C-Vis., and D-Vis. denote semantic, CLIP-visual, and DINOv2-visual branches; the bottom row is the full TRIC backbone.

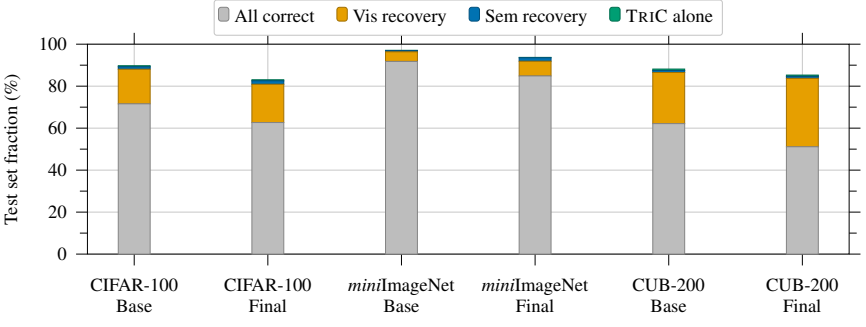



Figure 3: Per-query behavior of TRIC's inter-modal calibration at base and final sessions. Each stacked bar decomposes TRIC's correct predictions by how the two upstream branches voted: *all correct* (semantic, calibrated visual, and TRIC all right), *vis recovery* (semantic wrong, visual right, TRIC right), *sem recovery* (visual wrong, semantic right, TRIC right), and TRIC *alone* (only TRIC right). The *all-correct* band (gray) is the dominant contribution on every dataset; on the disagreement subset, TRIC defers to whichever single branch is right, and that branch shifts from semantic-supported at base to visual-driven at the final session as novel classes accumulate.

## 4.3 Ablation Study

We organise the ablations around three questions: *(i)* does the tri-level calibration backbone work, *(ii)* does QPR add measurable value on top of it, and *(iii)* are the new hyperparameters we introduce robust around the deployed defaults.

**TRIC effectiveness.** Table 2 enumerates the valid combinations of TRIC's three branches on *mini*ImageNet. Three contrasts carry the argument. (i) DINOv2 alone (Avg. 93.94, PD 4.52) exceeds CLIP-visual alone (Avg. 88.92, PD 5.79) by 5.02 Avg. points, identifying DINOv2 as the dominant low-shot novel carrier. (ii) The (Sem. + C-Vis.) row which is same as BiMC, lands at Avg. 92.97, already below DINOv2 alone, so the gain over comes from introducing the second visual encoder rather than from further calibration on top of CLIP. (iii) Bringing the semantic branch in on top of the calibrated visual one yields the full TRIC backbone at Avg. 95.65, PD 2.91, the largest single jump in the table (+1.28 Avg., −1.41 PD) and the only configuration to drop PD below 3.

Fig. 3 shows what the inter-modal calibration in Eq. 8 is actually doing per query. The

| Backbone | QPR | CIFAR-100 | | | *mini*ImageNet | | | CUB-200 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Full | Base | Novel | Full | Base | Novel | Full | Base | Novel |
| BiMC | off | 72.61 | 75.21 | 68.70 | 91.88 | 93.15 | 89.98 | 72.27 | 80.08 | 64.63 |
| BiMC | on | 73.50 | 74.65 | 71.78 | 92.31 | 92.92 | 91.39 | 73.50 | 79.55 | 67.59 |
| Δ | | +0.89 | −0.56 | +3.08 | +0.43 | −0.23 | +1.41 | +1.23 | −0.53 | +2.96 |
| BiMC + Ensemble | off | 73.24 | 77.79 | 66.41 | 92.41 | 92.97 | 91.57 | 72.76 | 78.89 | 66.77 |
| BiMC + Ensemble | on | 73.28 | 76.73 | 68.10 | 92.55 | 92.68 | 92.37 | 73.01 | 77.70 | 68.42 |
| Δ | | +0.04 | −1.06 | +1.69 | +0.14 | −0.29 | +0.80 | +0.25 | −1.19 | +1.65 |
| TRIC | off | 83.07 | 88.48 | 74.96 | 93.73 | 96.13 | 90.12 | 85.29 | 87.84 | 82.80 |
| TRIC | on | 85.50 | 88.01 | 81.75 | 94.74 | 95.80 | 93.15 | 86.17 | 87.72 | 84.66 |
| Δ | | +2.43 | −0.47 | +6.79 | +1.01 | −0.33 | +3.03 | +0.88 | −0.12 | +1.86 |

Table 3: Effect of adding QPR on top of three calibrated frozen-FSCIL backbones — plain BiMC, BiMC with its auxiliary ensemble, and TRIC — at the deployed default $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$, decomposed into Full / Base / Novel final-session accuracy. The Δ rows report the per-column gain over the corresponding pre-QPR row.

dominant contribution on every dataset is the *all-correct* band (gray): semantic and calibrated visual already agree on the same class, and TRIC reinforces that consensus into a sharper posterior. This is where the semantic branch carries its weight, not by overriding visual on disagreements, but by tightening confidence on agreements, which is what the trilevel ablation's +1.28 Avg. gain from adding the semantic branch on top of the calibrated visual one reflects. On the disagreement subset, TRIC defers to whichever branch is right: the *vis recovery* band (orange) grows from 16.48→18.23% on CIFAR-100, 4.58→7.00% on *mini*ImageNet, and 24.37→32.60% on CUB-200 between the base and final sessions, while the small *sem recovery* band (blue) remains nonzero, confirming that semantic dissent can still override visual when visual is wrong. The deferral rate therefore shifts from semantic-supported to visual-driven as novel classes accumulate, matching the trilevel ablation's identification of DINOv2 as the dominant low-shot novel carrier.

**QPR effectiveness.** Table 3 compares pre-QPR and post-QPR pipelines on three calibrated frozen-FSCIL backbones: plain BiMC, BiMC with its auxiliary ensemble, and TRIC — at the deployed default $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$, decomposed into Full / Base / Novel final-session accuracy.

QPR is a general label-free rectification stage, but its gain is set by the cleanness of the calibrated posterior the gate consumes. On plain BiMC (*mini*ImageNet), QPR adds +0.43 Full accuracy (91.88 → 92.31). On BiMC's full ensemble configuration the gain shrinks to +0.14 (92.41 → 92.55), because BiMC's description-nearest-neighbour branch already extracts query-side novel evidence at the description-retrieval stage; once that signal is consumed, the rectification head has little left to add. On TRIC, the same rectification delivers +1.01 Full and a sharply concentrated +3.03 Novel gain on *mini*ImageNet (and +6.79 / +1.86 Novel on CIFAR-100 and CUB-200), while *base* accuracy moves by at most 0.47 points in any direction. This ordering — bigger gain on cleaner posteriors — supports the structural claim that the tri-level calibration produces the cleanest posterior for the gate to act on, and that the gate then targets the noisy five-shot novel prototypes without leaking into base classes. Table 4 reports the gate diagnostics that make this safety possible on TRIC: purity is $\geq 97.79\%$ on every benchmark, base leakage stays $\leq 2.21\%$, and the admission rule successfully updates $40/40$, $40/40$, and $99/100$ novel classes.

| Dataset | Gate purity | Base leakage | Gated queries | Updated novel classes |
|---|---|---|---|---|
| CIFAR-100 | 97.79% | 2.21% | 3191.7±27.8 | 40/40 |
| *mini*ImageNet | 98.57% | 1.43% | 3743.1±25.8 | 40/40 |
| CUB-200 | 99.38% | 0.62% | 2861.2±8.0 | 99/100 |

Table 4: QPR final-session gate diagnostics (TRIC backbone, deployed configuration). Gate purity is the fraction of gated queries whose true class is novel; base leakage is the complement.

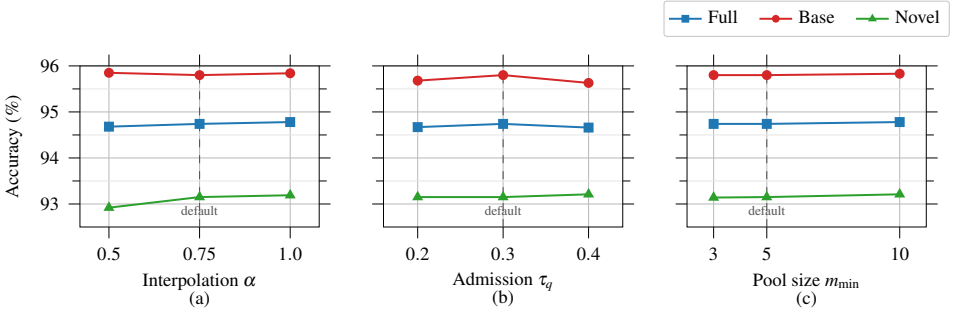



Figure 4: Sensitivity of the three QPR hyperparameters on *mini*ImageNet, varied one at a time around the deployed defaults $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$. The dashed vertical line marks the default in each panel; all panels share the same $y$-axis range to make the cross-sweep flatness directly visible.

The gate is not perfect: a small per-class diagnostic in the supplementary material shows that on fine-grained CUB the gate can occasionally route admitted queries to the wrong novel class and starve a competitor, which we discuss together with concrete extensions there.

**Hyperparameter sensitivity.** Fig. 4 sweeps the three QPR hyperparameters around the deployed defaults $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$ on *mini*ImageNet, varying one at a time. Across $\alpha \in \{0.5, 0.75, 1.0\}$, Full accuracy varies by only 0.10 points, Novel accuracy by only 0.27 points, and Base accuracy by only 0.05 points; the rectification result is therefore robust to the interpolation strength in the deployable range. The $\tau_q$ and $m_{\min}$ panels extend the sweep to the gate's admission threshold and the minimum admitted-pool size and show similarly small variation. The shared $y$-axis makes the across-sweep flatness directly visible: all three curves stay inside a window narrower than 0.3 points.

# 5 Conclusion

The largest single gain in our framework comes from the inter-model visual calibration in Eq. 7: DINOv2 prototypes calibrated against CLIP visual logits on the class-score axis carry most of the improvement over the BiMC baseline, and QPR then targets the residual five-shot prototype noise with unlabeled-query supervision through a conservative gate. Once a second visual encoder is introduced, frozen FSCIL is not one calibration but three, performed in *supervision-strength* order: the labeled-support-only operations (TRIC) come before the unlabeled-query operation (QPR), so that the gate QPR consumes is shaped by a posterior

the gate did not produce. TRIC-QPR uses no test labels at any stage and no per-seed tuning, making it directly applicable to frozen FSCIL systems without additional annotation overhead.

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